# HybridQPP: Scaling Synthetic Evaluation Signals through Pointwise and Pairwise LLM Alignment

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## Abstract

Advancing Information Retrieval (IR) systems requires scalable automated evaluation to bypass costly human or A/B testing. Query Performance Prediction (QPP) provides a fast, cost-effective automated alternative for assessing retrieval quality. Modern QPP frameworks have evolved to use Large Language Models (LLMs) as evaluators (LLM-as-a-Judge), typically employing one of two paradigms: pointwise or pairwise. While pointwise methods provide robust absolute measures and pairwise approaches offer comparative nuance, both suffer from performance volatility.

To overcome this problem, we propose HybridQPP, a novel QPP framework that bridges both paradigms within the synthetic IR lifecycle that combines pointwise and pairwise synthetic assessments. We demonstrate that this hybrid approach significantly outperforms pointwise and pairwise baselines, achieving superior Pearson's correlation on the TREC 2020/2022 datasets. Furthermore, HybridQPP ensures consistent stability by avoiding the severe performance drops often observed in individual methods, effectively minimizing prediction losses across all evaluated scenarios. Our findings contribute to the evaluation validity and practical deployment of LLM-driven assessment in next-generation retrieval ecosystems.

## CCS Concepts

• **Information systems → Evaluation of retrieval results**; **Retrieval effectiveness**; *Relevance assessment*.

## Keywords

Query performance prediction, Large language models, Synthetic Evaluation Validity

**ACM Reference Format:**
Cezary Golecki, Aleksander Wawer, and Paweł Zawistowski. 2026. HybridQPP: Scaling Synthetic Evaluation Signals through Pointwise and Pairwise LLM Alignment. In *Proceedings of The First Workshop on Synthetic Content in Information Retrieval Ecosystems (SIGIR 2026) (SynthIR 2026).* ACM, New York, NY, USA, 5 pages. https://doi.org/XXXXXXX.XXXXXXX


*SynthIR 2026, Melbourne, Australia*

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

## 1 Introduction

Query Performance Prediction is a long-standing and critical challenge in Information Retrieval, where, in a standard setting, a retriever processes a user query $q$ against a large corpus $D$ to produce a ranked list of items. The task of post-retrieval [3] Query Performance Prediction is to estimate the quality of this ranking, traditionally measured by metrics such as Reciprocal Rank (RR) or Normalized Discounted Cumulative Gain (nDCG), without access to ground-truth relevance labels.

Recent research has revealed the great potential of Large Language Models (LLMs) for automatically generating query performance predictions [9, 15]. This approach can lead to significant improvements compared to traditional QPP baselines. LLM-based methods typically use a pointwise or pairwise relevance judgment paradigm.

Following the terminology of the TREC Deep Learning (TREC-DL) track, we use the terms *question* and *passage* to refer to the query $q$ and document $d_i$, respectively.

The pointwise approach decomposes QPP into predicting the absolute relevance of each document independently [15]. This method enables direct calculation of IR metrics (e.g., nDCG@k, Precision@k) from the generated pseudo-labels. We use the following prompt to generate the pointwise annotations: ***Instruction****: Please assess the relevance of the provided passage to the following question. Please output "relevant" or "irrelevant".* ***Question****:* $q$. ***Passage****:* $d_i$. ***Output****:"*

Conversely, the pairwise approach integrates comparative analysis, prompting the LLM to judge the relative preference between two documents for a given query. Pairwise relevance judgments may enable a more nuanced assessment of ranking quality by capturing the relative context typically lost in the pointwise method. However, implementing the pairwise approach in QPP requires significantly more LLM inferences than the pointwise method. We use the following prompt to generate pairwise annotations: ***Instruction****: Please determine which of the two provided passages is more relevant to a given question. Please output "first", "second".* ***Question****:* $q$. ***First passage****:* $d_i$. ***Second passage****:* $d_j$. ***Output****:"*

As it is intuitive how to calculate IR metrics given pointwise relevance labels, it is not as clear given pairwise comparisons. [9] introduced several pairwise-comparison aggregation approaches, namely AllPairs, ReferencePairs, BubbleSort, and HeapSort. The AllPairs method involves an exhaustive comparison of every possible document pair within the retrieved list $\mathcal{N}$. The ReferencePairs method selects a subset of $K$ reference items from the document list and compares all other documents against them. In the BubbleSort and HeapSort approaches, the LLM serves as the comparison operator within the standard algorithms, and the comparisons generated during sorting are used to compute the final document's relevance score.

While both the pointwise and pairwise judgment paradigms have been shown to successfully estimate a QPP score, we demonstrate that their relative effectiveness may vary depending on the dataset and retrieval model used. Consequently, in this study, we introduce a HybridQPP framework that effectively combines the two approaches, integrating their complementary strengths and providing greater stability, while improving the performance in some cases.

## 2 Related Work

*Query Performance Prediction.* The QPP enables IR system evaluation without the need for expensive human annotations or online experiments. Post-retrieval QPP methods [3] offer diverse approaches, based on retriever scores [6, 19, 21, 22, 26, 31], comparing rankings across models [23], and utilizing dense retriever characteristics [1] and BERT embeddings [2, 7, 11, 13]. Recent research focuses on LLM-based relevance assessments [9, 15].

*LLMs for relevance judgments and ranking.* LLMs serve as an emerging alternative for generating relevance judgments. Pointwise prompting assigns labels to query-document pairs for direct computation of IR metrics, as seen in the QPP-GenRE framework [15]. Conversely, pairwise prompting (e.g., PRP) compares document pairs and can be adapted from ranking tasks to QPP [9]. Beyond ranking, LLMs supplement judgments [8] in conversational [16] and product-search relevance annotation [18, 25]. While some studies suggest LLMs approximate human quality [27], others disagree [24] or highlight inherent model inconsistencies and biases [20, 28]. It is also worth noticing that replacing humans in the evaluation loop could potentially save weeks and thousands of dollars per evaluation task [12], as human annotators are extremely costly.

## 3 HybridQPP

In Section 5, we show that QPP approaches based on either pointwise or pairwise paradigms demonstrate satisfactory correlations with human-based metrics. Yet they tend to excel in specific scenarios rather than deliver consistent performance across all contexts. To address this volatility, we introduce HybridQPP, a general framework that synthesizes pointwise and pairwise annotation paradigms to improve prediction robustness. We propose two integration strategies described below.

*HybridQPP$_{Agg}$.* This strategy involves directly combining the scalar scores generated by the pointwise (e.g., QPP-GenRE) and pairwise components. We employ four aggregation functions to fuse these signals: max, min, mean, and product. We denote the resulting methods as HybridQPP$_{\text{max}}$, HybridQPP$_{\text{min}}$, HybridQPP$_{\text{mean}}$, and HybridQPP$_{\text{prod}}$, respectively.

*HybridQPP$_{PoR}$.* Pairwise on Relevant (PoR) strategy utilizes a two-stage process. First, the pointwise model evaluates the retrieved list $\mathcal{D}_N$ to assign binary relevance labels. Second, the pairwise QPP score is computed solely over the subset of documents predicted as relevant. This approach focuses the pairwise comparison on potentially useful documents, reducing noise from irrelevant items.

## 4 Experiment Setup

Following Meng et al. [15] and Golecki et al. [9], we replicate the experimental setup for evaluating LLMs on IR tasks. We use the TREC Deep Learning 2020 (TREC'20) and 2022 (TREC'22) tracks [4, 5] as our primary benchmarks, utilizing the MS MARCO V1 and V2 [17] collections respectively. Document retrieval is performed using Pyserini [14] with BM25 ($k_1 = 0.9, b = 0.4$) and ANCE [29] dense retrieval.

Our evaluation employs Llama-3.1-8B-Instruct [10] and Qwen3-8B [30]. We use both zero-shot base models and versions fine-tuned on TREC'19, following the procedure described in Golecki et al. [9]. All inferences are conducted at a temperature of 0. Consistent with prior work, we distinguish between retrieval depths for different paradigms, setting $N_{\text{pointwise}} = 200$ and $N_{\text{pairwise}} = 80$, and for pairwise methods—including RefPairs, BubbleSort, and HeapSort—we use $k = 7$; these parameters represent a reduction from Golecki et al. [9] to optimize inference costs without significant performance degradation (we provide parameter sensitivity analysis in Section 5). Performance is measured via correlation coefficients between human-annotated NDCG scores and LLM-generated signals. Note that we do not report RR, following the discussion from Golecki et al. [9].

## 5 Results

We present an empirical evaluation of our proposed HybridQPP framework. First, we analyze the aggregation function used to combine distinct signal types. Then, we point out the performance volatility of pointwise and pairwise approaches. Further, we benchmark HybridQPP against strong pointwise and pairwise baselines in terms of reduced performance volatility and improved correlation with the ground truth. We did not include the non-LLM baselines in the results, as they are available in [15] and directly comparable to our results. Also note that the goal of this paper is to improve known LLM-based QPP approaches, which already outperform the non-LLM baselines. Finally, we analyze the computational trade-offs between predictive robustness and the number of LLM inferences required.

*Analysis of Aggregation Methods in HybridQPP$_{Agg}$.* Table 1 reports the mean Pearson ($\rho$) and Kendall's ($\tau$) correlation coefficients across all 3 datasets and retriever setups. The multiplicative strategy, HybridQPP$_{\text{prod}}$, achieves the highest overall performance, notably for NDCG@10 $\rho$ (0.54 ± 0.09). The mean strategy is highly competitive and lost with the product by a narrow margin. In contrast,

**Table 1: Mean NDCG@10 across pairwise strategies and all datasets-retriever setups, ± standard deviation**

| QPP approach | NDCG@10 $\rho$ | NDCG@10 $\tau$ |
|---|---|---|
| HybridQPP$_{\text{Mean}}$ | 0.53 ± 0.09 | 0.37 ± 0.07 |
| HybridQPP$_{\text{Max}}$ | 0.44 ± 0.11 | 0.30 ± 0.07 |
| HybridQPP$_{\text{Min}}$ | 0.50 ± 0.01 | 0.34 ± 0.08 |
| HybridQPP$_{\text{Product}}$ | 0.54 ± 0.09 | 0.37 ± 0.07 |

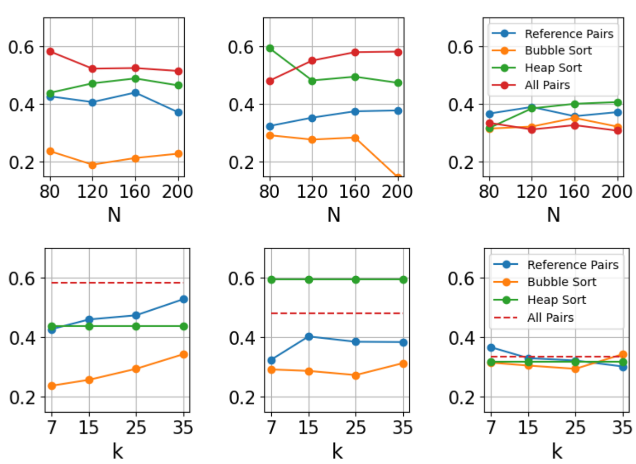
strategies utilizing extreme values (Max and Min) consistently underperform, suggesting they fail to properly integrate signals of varying strengths. Consequently, we select product as the aggregation function for HybridQPP$_{\text{Agg}}$.

*Instability of Pointwise and Pairwise QPP Methods.* Results show neither the pointwise nor the pairwise paradigm is universally superior. As in Table 2, the relative merits of these two methodologies are extremely volatile and tend to change dramatically depending on the LLM used, the retrieval dataset, and the model's state (Base vs. Fine-Tuned). Take the case of AllPairs using Qwen3 (Base) on the TREC'20 (ANCE) dataset; here, QPP-GenRe using the Pointwise approach simply collapses, posting a correlation deficit of −74.8% on its corresponding Pairwise baseline of 0.538. Conversely, this trend is completely reversed when testing Qwen3 (Fine-Tuned) on TREC'22; here, the Pairwise AllPairs methodology underperforms the Pointwise baseline ($\rho = 0.505$) by as much as −68.6%. On Llama-3.1-8B TREC'20 (BM25), the Base model favors Pairwise (AllPairs baseline $\rho = 0.481$), with Pointwise lagging by −17.6%. Conversely, Fine-Tuning shifts preference to Pointwise QPP-GenRe ($\rho = 0.759$), causing a drastic swing to a deficit of −32.6% from the previously favored Pairwise method. The different sorting methods used to guide performance measurement within the Pairwise model, BubbleSort, and HeapSort, have shown similar volatile performance measurement as well. For example, on TREC'20 (ANCE), BubbleSort performed 49.3% worse than HeapSort in the same dataset-retriever setup, but the trend reverses when looking at the TREC'22 (BM25). These performance troughs underscore the risk of single-paradigm selection and the necessity of HybridQPP to mitigate these failures.

*Effectiveness and stability of HybridQPP.* HybridQPP$_{\text{prod}}$ demonstrates consistency and high peak performance, especially in BubbleSort, where it achieves a relative improvement of +37.2% for the base Qwen on TREC'20 with ANCE retriever. BubbleSort with base Qwen delivers a significant improvement of +22.7% over the performance of the individual baselines. For TREC'22 with BM25, the largest gains are achieved by HeapSort with base Llama and RefPairs with fine-tuned Llama, achieving 16.8% and 15.5%, respectively.

HybridQPP$_{\text{PoR}}$ consistently outperforms the baseline methods. On TREC'20 with the ANCE retriever, the best result was achieved by fine-tuned Qwen combined with BubbleSort, yielding a +39.1% improvement. For TREC'20 using BM25, HybridQPP$_{\text{PoR}}$ resulted in gains of +25.1% with base Llama and +5.0% with base Qwen, both using BubbleSort. On TREC'22 (BM25), base Llama with AllPairs delivered the largest improvement over the baselines – +16.7%.

Table 2 highlights HybridQPP's stability and risk mitigation properties. Considering the "Min" column, we observe that although individual baselines are sometimes subject to catastrophic degradation, such as the −74.8% drop in AllPairs (QPP-GenRe) or the −73.6% drop in HeapSort, the HybridQPP methods drastically dampen these worst-case outcomes. In RefPairs, HybridQPP$_{\text{prod}}$ limits the worst-case drop to no more than −3.7% opposed to −71.1% for QPP-GenRe. Furthermore, HybridQPP maintains positive or near-zero mean performance, whereas constituent baselines often exhibit negative averages.




**(a) TREC'20, ANCE** **(b) TREC'20, BM25** **(c) TREC'22, BM25**

**Figure 1: Pearson's correlation coefficients for different pairwise QPP approaches with different values of parameters $N_{\text{pairwise}}$ (top) and $k$ (bottom).**

*Inference Counts.* Evaluating the "# calls" column in Table 2, the HybridQPP$_{\text{prod}}$ approach always adds a fixed overhead of 200 inferences for all sorting variants over the standalone Pairwise approach. In the RefPairs setup, for example, the approach always incurs 760 inferences as opposed to 560 of the standalone Pairwise approach. Although the overhead is 36%, it must be accounted for when using the HybridQPP approach to achieve the robustness referenced in the previous paragraph. On the other hand, the HybridQPP$_{\text{PoR}}$ (Pairwise on Relevant) strategy takes advantage of the Pointwise information to prune the number of pairwise inference calls dynamically. This optimization benefits every sorting algorithm and is crucial for complex scenarios. Under the AllPairs setting, previously known to be 6400 inference calls per query, the PoR strategy achieves an average of 3735 inference calls per query. This represents a decrease of roughly 42% and makes the application of the high-fidelity AllPairs signals more plausible in certain resource-constrained scenarios. Even with efficient sorting like HeapSort, PoR provides clear benefits. HybridQPP$_{\text{PoR}}$ averages 347 inferences, a 16% reduction over the HybridQPP$_{\text{prod}}$ (413), confirming that its pruning enhances efficiency without sacrificing accuracy. Please note that the QPP methods serve as an offline retriever/ranker evaluator, so despite the large magnitude of LLM calls, calculating the presented QPPs is feasible.

*Hyperparameter $k$ Sensitivity Study.* Figure 1 justifies our pairwise QPP parameter adjustments relative to [9]. Subfigures show that pairwise performance is stable across different values of $k$. For instance, in both TREC'20/22 BM25, Pearson's correlation coefficients for Heap Sort and the All Pairs baseline remain horizontal, indicating that the predictive power is independent of the document neighborhood size. Fluctuations in Reference Pairs and Bubble Sort (e.g., TREC'20 ANCE) are marginal and show no significant correlation between hyperparameters and accuracy. The relative ranking of the different strategies remains largely unchanged regardless of whether $k$ is set to 7 or 35, suggesting that LLM-based pairwise signals are robust and require minimal tuning.

**Table 2: Correlation coefficients (Pearson's $\rho$) between different QPP methods and NDCG@10 calculated using human generated labels. The best correlation for a dataset-model pair is bold, the worst is underlined. * indicates statistically significant estimates (p value $< 0.05$ - two-sided Student's t-test under the null hypothesis of zero correlation). The "Aggregated Statistics" section reports the percentage difference between the minimum, mean, and maximum correlation values in the corresponding row and the best corresponding non-hybrid method (i.e., pointwise or one of AllPairs, RefPairs, BubbleSort, HeapSort). The "# calls" column indicates the average number of LLM inference calls (across models and their variants) required to obtain each result.**

| Setup | TREC'20 (ANCE) | | | | TREC'20 (BM25) | | | | TREC'22 (BM25) | | | | Aggregated Statistics | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Llama | | Qwen | | Llama | | Qwen | | Llama | | Qwen | | Min | | Mean | | Max | | Avg |
| Ver. | Base | FT | Base | FT | Base | FT | Base | FT | Base | FT | Base | FT | Base | FT | Base | FT | Base | FT | # calls |
| AllPairs | | | | | | | | | | | | | | | | | | | |
| QPP-GenRe | 0.464* | 0.363* | 0.136 | 0.379* | 0.396* | 0.759* | 0.246 | 0.724* | 0.435* | 0.463* | 0.338* | 0.505* | -74.8% | -39.0% | -16.4% | -5.5% | 0.0% | 0.0% | 200 |
| AllPairs | $\mathbf{0.584}^*$ | $\mathbf{0.594}^*$ | $\mathbf{0.538}^*$ | 0.522* | 0.481* | 0.512* | $\mathbf{0.511}^*$ | 0.403* | 0.335* | 0.326* | 0.502* | 0.158* | -23.0% | -68.6% | 0.0% | -14.2% | 0.0% | 0.0% | 6400 |
| $\text{HybridQPP}_{\text{prod}}$ | 0.511* | 0.534* | 0.478* | 0.547* | 0.442* | 0.772* | 0.464* | 0.733* | 0.434* | 0.485* | 0.515* | 0.525* | -12.4% | -18.4% | -2.3% | -0.2% | 11.1% | 4.8% | 6600 |
| $\text{HybridQPP}_{\text{PoR}}$ | 0.533* | 0.485* | 0.526* | $\mathbf{0.609}^*$ | 0.472* | 0.745* | 0.467* | 0.779* | $\mathbf{0.508}^*$ | 0.521* | 0.548* | 0.533* | -8.7% | -10.1% | 0.4% | 2.5% | 16.7% | 16.6% | 3735 |
| RefPairs | | | | | | | | | | | | | | | | | | | |
| QPP-GenRe | 0.464* | 0.363* | 0.136 | 0.379* | 0.396* | 0.759* | 0.246 | 0.724* | 0.435* | 0.463* | 0.338* | 0.505* | -71.1% | -12.9% | -12.6% | -1.1% | 0.0% | 0.0% | 200 |
| RefPairs | 0.427* | 0.334* | 0.470* | 0.436* | 0.324* | 0.615* | 0.420* | 0.402* | 0.367* | 0.436* | 0.547* | 0.348* | -18.2% | -44.5% | -0.4% | -9.0% | 0.0% | 0.0% | 560 |
| $\text{HybridQPP}_{\text{prod}}$ | 0.512* | 0.456* | 0.452* | 0.556* | 0.430* | **0.785** | 0.444* | 0.723* | 0.493* | $\mathbf{0.535}^*$ | 0.557* | $\mathbf{0.571}^*$ | -3.7% | 0.1% | 3.0% | 7.1% | 13.2% | 27.7% | 760 |
| $\text{HybridQPP}_{\text{PoR}}$ | 0.430* | 0.393* | 0.462* | 0.474* | 0.415* | 0.710 | 0.470* | 0.709* | 0.497* | 0.506* | 0.534* | 0.535* | -7.2% | -6.5% | 1.6% | 2.0% | 14.1% | 9.3% | 564 |
| BubbleSort | | | | | | | | | | | | | | | | | | | |
| QPP-GenRe | 0.464* | 0.363* | 0.136 | 0.379* | 0.396* | 0.759* | 0.246 | 0.724* | 0.435* | 0.463* | 0.338* | 0.505* | -48.0% | -5.0% | -6.1% | -0.4% | 0.0% | 0.0% | 200 |
| BubbleSort | 0.237 | 0.322* | 0.261 | 0.376* | 0.292* | 0.464* | 0.207 | 0.383* | 0.315* | 0.487* | 0.450* | 0.173 | -48.9% | -65.8% | -9.9% | -13.7% | 0.0% | 0.0% | 532 |
| $\text{HybridQPP}_{\text{prod}}$ | 0.458* | 0.440* | 0.296* | 0.521* | 0.477* | 0.745* | 0.302* | 0.730* | 0.468* | 0.528* | 0.509* | 0.541* | -1.3% | -1.9% | 6.2% | 5.3% | 22.7% | 37.2% | 732 |
| $\text{HybridQPP}_{\text{PoR}}$ | 0.359* | 0.416* | 0.314* | 0.528* | 0.416* | 0.694* | 0.307* | 0.682* | 0.436* | 0.490* | $\mathbf{0.564}^*$ | 0.515* | -22.6% | -8.6% | 4.2% | 4.1% | 25.3% | 39.1% | 554 |
| HeapSort | | | | | | | | | | | | | | | | | | | |
| QPP-GenRe | 0.464* | 0.363* | 0.136 | 0.379* | 0.396* | 0.759* | 0.246 | 0.724* | 0.435* | 0.463* | 0.338* | 0.505* | -73.63% | -33.3% | -15.0% | -5.4% | 0.0% | 0.0% | 200 |
| HeapSort | 0.439* | 0.508* | 0.515* | 0.598* | $\mathbf{0.594}^*$ | 0.466* | 0.445* | 0.436* | 0.317* | 0.293* | 0.470* | 0.181 | -27.1% | -39.7% | -2.7% | -14.9% | 0.0% | 0.0% | 213 |
| $\text{HybridQPP}_{\text{prod}}$ | 0.456* | 0.523* | 0.486* | 0.607* | 0.536* | 0.731* | 0.442* | 0.713* | $\mathbf{0.508}^*$ | 0.456* | 0.492* | 0.564* | -9.8% | -3.7% | 0.3% | 0.8% | 16.8% | 11.7% | 413 |
| $\text{HybridQPP}_{\text{PoR}}$ | 0.582* | 0.521* | 0.528* | 0.601* | 0.492* | 0.686* | 0.480* | $\mathbf{0.812}^*$ | 0.506* | 0.359* | 0.476* | 0.521* | -17.2% | -22.5% | 3.0% | -1.1% | 25.5% | 12.2% | 347 |

*Hyperparameter $N_{pairwise}$ Sensitivity Study.* Furthermore, we examine the sensitivity of our approach to the number of documents considered for pairwise comparisons, denoted as $N_{\text{pairwise}}$. As shown in Figure 1, increasing $N_{\text{pairwise}}$ from 80 to 200 does not yield a clear or consistent improvement in Pearson's correlation coefficients across datasets and retrieval models. In most cases, such as with Heap Sort and Reference Pairs in the TREC'22 BM25 setting, the performance remains relatively flat or exhibits minor, non-monotonic fluctuations that do not suggest a significant dependency on the parameter scale. While some methods show localized variance—for example, the performance of BubbleSort on TREC'22 or the AllPairs baseline on TREC'20 — these shifts are inconsistent across different settings and do not demonstrate a scalable benefit to using higher values of $N_{\text{pairwise}}$. The stability of these results indicates that the pairwise QPP signals are sufficiently captured even at lower values of $N$, suggesting that the quality prediction performance is robust to this hyperparameter. This lack of a strong correlation justifies our selection of a fixed $N_{\text{pairwise}}$ value, balancing predictive accuracy with the computational cost of evaluating larger document sets.

## 6 Conclusions

In this paper, we investigated the system-level implications of utilizing LLMs as synthetic evaluators for QPP. Our work addresses a critical challenge in modern IR ecosystems: the need for valid, automated assessment signals when human relevance labels are unavailable or impractical to scale. We demonstrated that the current state-of-the-art approaches are unstable and sometimes critically underperform.

To ensure evaluation validity across multiple datasets, we introduced HybridQPP, a framework that integrates the strengths of both annotation paradigms. Our experiments on the TREC 2020 and 2022 benchmarks confirm that HybridQPP significantly outperforms individual pointwise and pairwise approaches, while keeping any performance degradation to a minimum. Our method achieved a higher Pearson correlation with human-label-derived NDCG than the individual pairwise or pointwise estimations. This suggests that the combinations of two different judges provide a more robust foundation for monitoring retrieval quality.

We explored hybrid approaches with $\text{HybridQPP}_{\text{prod}}$ with BubbleSort as the underlying pairwise QPP method, as the one with the highest average gain over the baselines. Its maximal gains are +22.7 and +39.1% for the base and finetuned models, respectively, while its maximum observed losses were negligible, at -1.3% and -1.9%. In comparison, the baselines' maximal losses vary from $-5.0\%$ to $-65.8\%$. These results underscore the importance of multifaceted assessment frameworks in building trustworthy, grounded IR evaluation systems.

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Received 18 December 2026; revised 12 March 2009; accepted 5 June 2009